

S&P 500 Sector Dynamics: Return Connectedness, Market Concentration, and Structural Clustering (2019 to 2025)

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Abstract

This working paper studies the structure and dynamics of the eleven GICS sectors of the S&P 500 over September 2019 to December 2025, combining time-series econometrics, industrial-organization measures, and unsupervised learning across 401 constituents. Using daily market-cap-weighted sector returns (1,590 observations), we estimate a Diebold-Yilmaz (2012) generalized connectedness system: the total connectedness index is 78.6%, indicating that most sector forecast-error variance is cross-sectional rather than own-sector. Industrials and Financials are the largest net transmitters of shocks while Energy is the largest net receiver, and a rolling 200-day connectedness index rises from about 55% to a 2020 peak of 87%, tracing systemic stress through the COVID-19 crisis. Pairwise Granger causality on the daily series, corrected for multiple testing, finds 88 of 110 directed links significant under Benjamini-Hochberg FDR (54 under the stricter Bonferroni); the same test on quarterly data is underpowered and yields no links surviving correction, underscoring the role of sampling frequency and multiple-testing control. At the sub-sector level (69 GICS sub-industries), market concentration measured by the Herfindahl-Hirschman Index is rising on net (45 of 69 sub-sectors concentrating), led by Semiconductors and Automobile Manufacturers. Hierarchical and K-means clustering recovers a coherent two-group sector taxonomy, and out-of-sample tests show that broad market-structure labels are fairly stable while growth-dominance rankings reshuffle ($r = 0.181$, $p = 0.14$), indicating that sector dominance is not persistent.

Keywords: S&P 500, sector connectedness, Diebold-Yilmaz, VAR, generalized variance decomposition, Granger causality, Herfindahl-Hirschman Index, market concentration, clustering, time-series econometrics

1 Introduction

Understanding competitive dynamics within stock market sectors is fundamental to investment strategy and regulatory policy [1, 13]. While individual stock analysis is well-studied, the sector-level perspective reveals structural patterns that are invisible at the company level: whether an industry is dominated by one firm, shared between two, or dispersed across many competitors.

Traditional industrial organization uses the Herfindahl-Hirschman Index (HHI) to measure market concentration based on revenue or market share [4]. However, static concentration metrics may

not capture dynamic competitive advantages. A firm can dominate its sector’s *growth* even if its absolute market share is moderate. This paper introduces an “overperformance index” that uses a squared-weight allocation to capture growth-based dominance, and validates it against the traditional HHI framework.

The analysis covers 68 GICS sub-sectors of the S&P 500 index with quarterly market capitalization data spanning September 2019 to December 2025 (26 quarters), and quarterly revenue data for the same period. Five normalized parameters are computed per sector and used for unsupervised clustering. The methodology extends prior work in four key dimensions:

1. **Statistical rigor:** Market structure classifications are supported by binomial significance tests, not arbitrary thresholds, with cross-validation through HHI comparison.
2. **Temporal out-of-sample validation:** An in-sample (2019-2023) / out-of-sample (2024-2025) split tests whether market structures and concentration rankings persist beyond the training window.
3. **Time-varying analysis:** Rolling-window overperformance concentration shows how market structures evolve across market cycles, rather than treating them as static.
4. **Cross-sector connectedness:** a Diebold-Yilmaz generalized variance-decomposition system on daily sector returns quantifies shock propagation across sectors (total connectedness, net transmitters and receivers, and a time-varying index), complemented by multiple-testing-corrected Granger causality.

2 Related Work

The application of clustering techniques to financial data has a rich history. Mantegna [10] introduced hierarchical clustering of stocks based on return correlations, establishing the concept of market taxonomy. Tumminello et al. [11] extended this to analyze the dynamic evolution of financial networks.

Market concentration measurement using HHI is standard in industrial economics [4]. The U.S. Department of Justice considers markets with HHI below 0.15 as unconcentrated, 0.15-0.25 as moderately concentrated, and above 0.25 as highly concentrated [5]. We adapt these thresholds for our overperformance-based HHI.

Granger causality has been applied to financial time series by Billio et al. [12], who used it to measure interconnectedness among financial institutions as a systemic risk indicator. Our application extends this to sector-level returns rather than individual institutions.

Beta estimation for sectors follows the capital asset pricing model framework [2], with the distinction that we compute market-capitalization-weighted sector returns as the dependent variable rather than individual stock returns.

3 Methodology

3.1 Data and Preprocessing

We collect quarterly market capitalization and quarterly revenue data for all S&P 500 constituent companies, grouped by GICS sub-industry classification. Market capitalization is computed as Close price $\times$ shares outstanding, sourced from Yahoo Finance. Sectors with fewer than 3 constituent companies are excluded to ensure statistical reliability. After filtering, the dataset comprises 68 qualifying sub-sectors spanning September 2019 through December 2025 (26 quarters).

3.2 Financial Metric Computation

Five parameters are computed for each sector.

3.2.1 Market Capitalization Growth Score

Year-over-year growth is computed as:

$$r_{i,t} = \frac{M_{i,t} - M_{i,t-4}}{M_{i,t-4}} \quad (1)$$

where $M_{i,t}$ is the market capitalization of company i at quarter t , and $t - 4$ denotes the same quarter one year prior. Growth is scored using a threshold of 0.5σ from the sector mean:

$$s_{i,t} = \begin{cases} +1 & \text{if } r_{i,t} > \mu + 0.5\sigma \\ -1 & \text{if } r_{i,t} < \mu - 0.5\sigma \\ 0 & \text{otherwise} \end{cases} \quad (2)$$

The 0.5σ threshold places approximately 38% of observations in the neutral band and 31% in each tail, creating meaningful differentiation between outperforming, underperforming, and neutral growth periods.

3.2.2 Revenue Growth Score

Computed identically to the market cap growth score but using quarterly revenue data, providing a fundamental (as opposed to market-based) measure of sector competitiveness.

3.2.3 Weighted-Simple Variance (Overperformance Index)

For each company i , let c_i be the count of quarters in which company i 's YoY market cap growth exceeded the sector mean. Define the squared-weight:

$$w_i = \frac{c_i^2}{\sum_j c_j^2} \quad (3)$$

The overperformance index is the mean absolute difference between the weighted average growth (using w_i) and the simple average growth across all quarters. The squaring in Eq. 3 amplifies the influence of consistently dominant firms, making the index sensitive to monopolistic structures. Large values indicate that a few firms disproportionately drive sector performance.

3.2.4 Beta Covariance

Sector beta measures sensitivity to market movements, computed as:

$$\beta = \frac{\text{Cov}(R_s, R_m)}{\text{Var}(R_m)} \quad (4)$$

where R_s is the market-capitalization-weighted sector return and R_m is the equal-weighted cross-sector return. We compute both short-term (2-quarter) and long-term (16-quarter) betas to capture different frequency co-movement patterns.

3.3 Market Structure Classification

3.3.1 Binomial Significance Test

Under the null hypothesis that outperformance is random (no persistent competitive advantage), the probability of beating the sector mean in any given quarter is $p = 0.5$. For each company, we test:

$$H_0 : p = 0.5 \quad \text{vs.} \quad H_1 : p > 0.5 \quad (5)$$

using the exact binomial test. A sector is classified as monopolistic if one company has a statistically significant ($\alpha = 0.05$) outperformance rate and captures $> 35\%$ of total outperformance counts.

3.3.2 HHI Validation

We compute HHI on overperformance counts:

$$\text{HHI}_{\text{overperf}} = \sum_i \left(\frac{c_i}{\sum_j c_j} \right)^2 \quad (6)$$

and compare against traditional HHI from market capitalization shares:

$$\text{HHI}_{\text{mktcap}} = \sum_i \left(\frac{M_i}{\sum_j M_j} \right)^2 \quad (7)$$

Classification thresholds follow DOJ guidelines [5]: $\text{HHI} > 0.25$ (monopolistic), $0.15\text{--}0.25$ (duopolistic), < 0.15 (oligopolistic).

3.4 Sector Clustering

The five normalized parameters are standardized using z-score normalization and clustered using a two-stage approach:

1. **Hierarchical clustering** with Ward's minimum variance linkage [6] to visualize cluster structure via dendrogram.
2. **K-Means clustering** with optimal k selected by maximizing the silhouette score [7] over $k \in \{2, \dots, 10\}$.

3.4.1 Validation

Rather than training a neural network on $N = 68$ samples (which would be statistically meaningless due to extreme overfitting risk), we validate cluster quality using:

- Silhouette score (cohesion vs. separation) [7]
- Calinski-Harabasz index (between- vs. within-cluster variance)
- Davies-Bouldin index (cluster compactness)
- Leave-one-out cross-validation with Random Forest [8], which provides a proper small-sample accuracy estimate without data leakage

3.5 In-Sample / Out-of-Sample Validation

To assess whether identified market structures are persistent or ephemeral, we split the data temporally:

- **In-sample:** September 2019 - December 2023 (17 quarters, 70%)
- **Out-of-sample:** January 2024 - December 2025 (8 quarters, 30%)

All metrics, overperformance rankings, and market structure classifications are computed independently on each split. We measure: (a) the Pearson correlation of overperformance index values between splits, and (b) the agreement rate of market structure classifications (monopolistic/duopolistic/oligopolistic).

3.6 Time-Varying Analysis

To assess whether market structures are stable or evolving, we compute the HHI of overperformance counts in a rolling 8-quarter (2-year) window for each sector. This reveals transitions between competitive regimes across different market cycles (pre-COVID, pandemic, recovery, and the 2024-2025 AI-driven rally).

3.7 Cross-Sector Connectedness and Granger Causality

Cross-sector dynamics are studied on *daily* market-cap-weighted sector returns (about 1,590 observations per series) rather than the 25 quarterly points used elsewhere, which are far too few for a VAR.

Generalized connectedness. We fit a VAR(p) on the 11 GICS-sector daily returns (p by AIC) and compute the generalized forecast-error variance decomposition (Pesaran and Shin, 1998) at a 10-day horizon, row-normalized into the Diebold and Yilmaz (2012) connectedness table, from which we report the total connectedness index, net directional connectedness (transmitters versus receivers), and a rolling 200-day total index.

Granger causality. We also test pairwise Granger causality [9]:

$$R_{i,t} = \alpha + \sum_{l=1}^L \beta_l R_{i,t-l} + \sum_{l=1}^L \gamma_l R_{j,t-l} + \varepsilon_t \quad (8)$$

with an F-test of $H_0 : \gamma_1 = \dots = \gamma_L = 0$ and maximum lag $L = 5$ days. Because all directed pairs are tested at once, p -values are corrected with both Benjamini-Hochberg FDR and Bonferroni, and we report survivor counts rather than the raw count.

4 Results

4.1 Market Structure Classification

The overperformance index analysis reveals heterogeneous competitive dynamics across the 68 sub-sectors.

Binomial test classification: 37 sectors (54.4%) exhibit oligopolistic environments, 22 sectors (32.4%) show monopolistic tendencies, and 9 sectors (13.2%) display duopolistic structures.

HHI-based classification: 33 sectors (48.5%) are classified as monopolistic, 18 (26.5%) as duopolistic, and 17 (25.0%) as oligopolistic.

The discrepancy between methods is informative: HHI computed from market capitalization shares reflects static size dominance, while the binomial test on outperformance counts captures *dynamic* growth dominance. Sectors where the two methods agree represent robust monopolistic structures; disagreements highlight sectors where size dominance and growth dominance diverge.

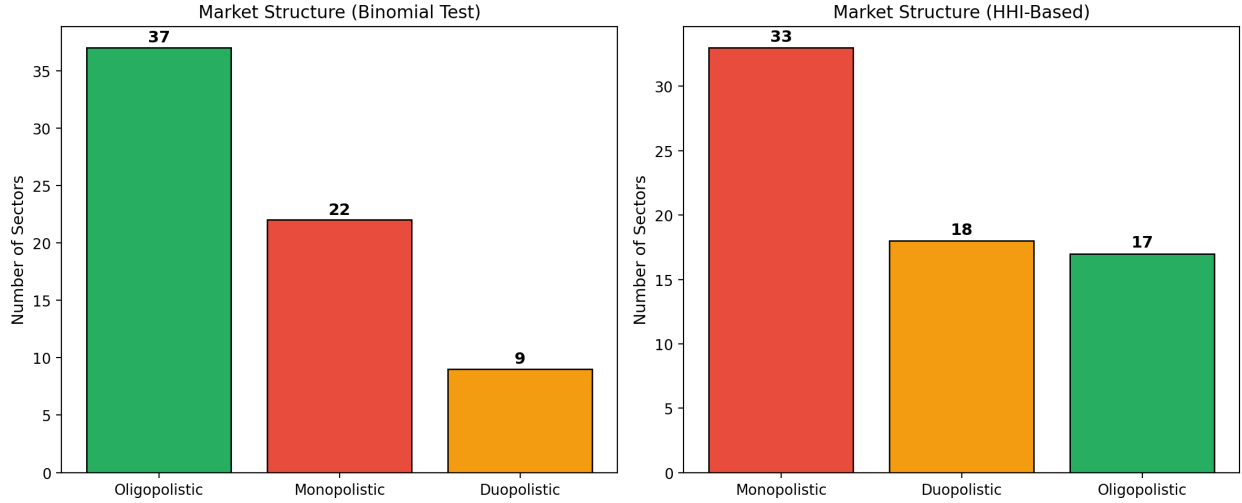


Figure 1: Market structure classification under two methods. The binomial test (left) identifies more oligopolistic sectors because it requires statistical significance, not just high concentration. The HHI method (right) finds more monopolistic sectors based on raw overperformance count shares.

Table 1 presents the sectors with the highest traditional HHI from market capitalization shares. Notably, Internet Services & Infrastructure (dominated by Palantir at 88.8%) has risen to the third-highest concentration, reflecting PLTR’s extraordinary market cap growth during 2024-2025.

Table 1: Most Concentrated Sectors by Market Cap HHI

Sector	Top Co.	Share	HHI _{MC}
Tech. Hardware, Storage	AAPL	94.1%	0.886
Automobile Mfrs.	TSLA	93.1%	0.869
Internet Svc. & Infra.	PLTR	88.8%	0.793
Systems Software	MSFT	87.7%	0.773
Personal Care Products	PG	85.0%	0.734

4.2 HHI Cross-Validation

The correlation between HHI computed from market cap shares and HHI from overperformance counts is $r = 0.537$ ($p < 10^{-4}$), indicating substantial but imperfect agreement. This validates the overperformance index as a meaningful concentration metric while also showing that it captures distinct information not present in static market shares.

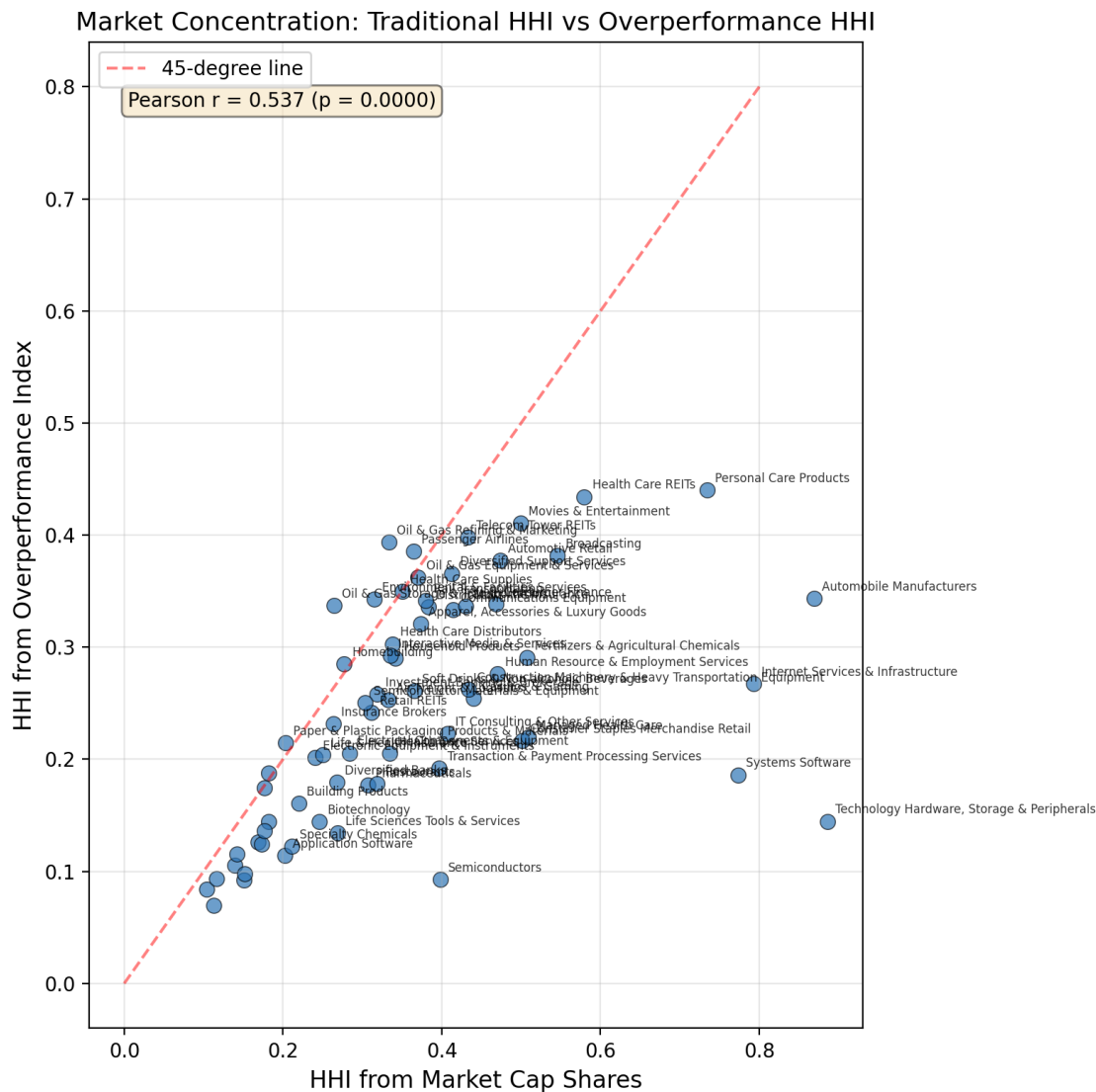


Figure 2: HHI from market capitalization shares vs. HHI from overperformance counts. The positive correlation ($r = 0.537$) validates the overperformance index, while scatter around the 45-degree line reveals sectors where size dominance and growth dominance diverge.

4.3 Sector Clustering

4.3.1 Optimal Cluster Selection

Silhouette analysis identifies $k = 2$ as the statistically optimal partition (score = 0.360), substantially above all alternatives ($k = 3$: 0.252; $k = 5$: 0.265; $k = 7$: 0.249). This indicates that the 68 sectors separate most cleanly into two groups: a large cluster of 55 sectors with moderate, dispersed competitive dynamics, and a smaller cluster of 13 sectors with extreme concentration or unusual financial metric profiles.

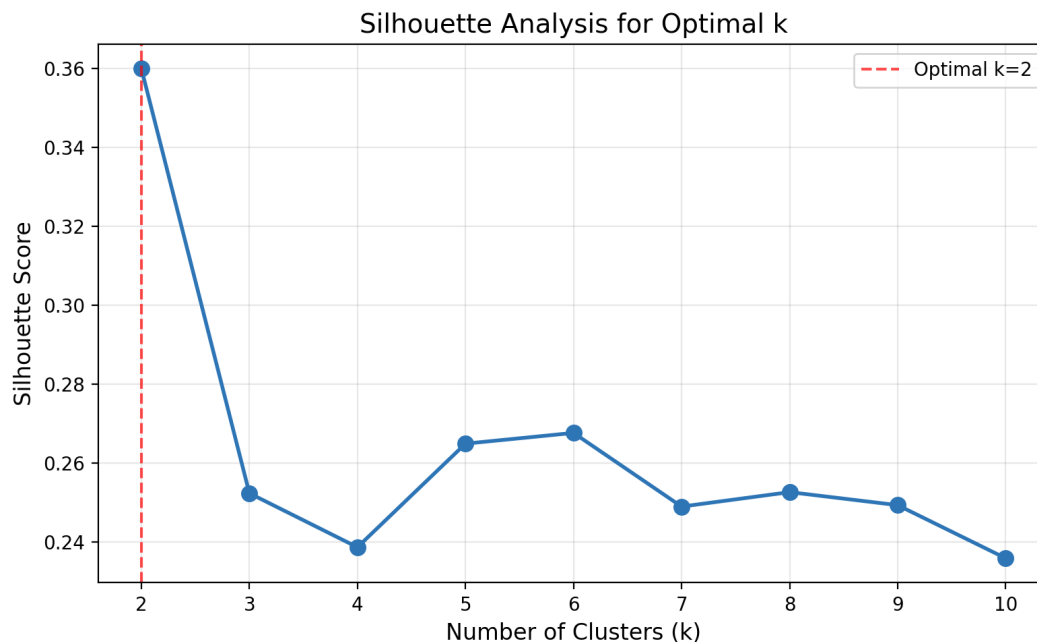


Figure 3: Silhouette scores for $k \in \{2, \dots, 10\}$. The dominant peak at $k = 2$ (score = 0.360) reflects a clean binary partition in the data. Secondary peaks at $k = 5$ -6 suggest finer sub-structure within the two main groups.

4.3.2 Cluster Composition

Cluster 0 (55 sectors) represents the “typical” competitive landscape: moderate growth variance, balanced overperformance distributions, and market-average beta exposures. This cluster includes sectors such as Packaged Foods & Meats, Aerospace & Defense, Diversified Banks, and Health Care Equipment.

Cluster 1 (13 sectors) captures outlier sectors with extreme concentration or unusual metric profiles. This cluster includes Technology Hardware (Apple-dominated), Automobile Manufacturers (Tesla-dominated), Internet Services & Infrastructure (Palantir-dominated), Semiconductors, and Communications Equipment. These sectors share a common characteristic: at least one company captures a disproportionate share of the sector’s growth trajectory.

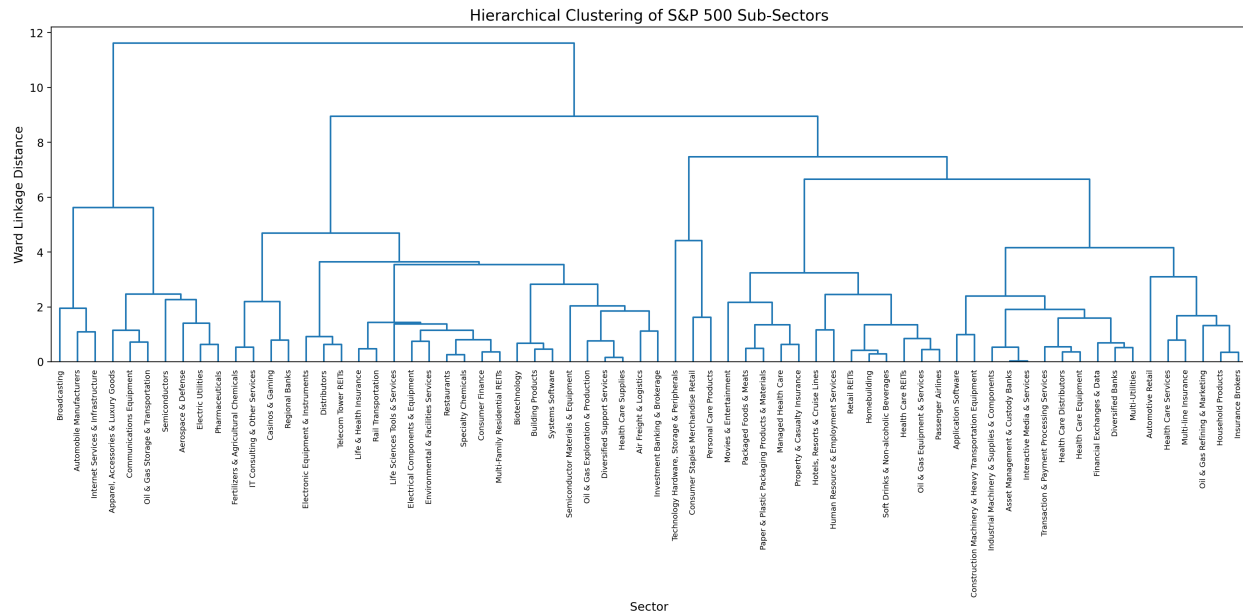


Figure 4: Dendrogram of hierarchical clustering (Ward's method) across 68 sub-sectors. Data through December 2025. The primary bifurcation separates high-concentration from moderate-concentration sectors.

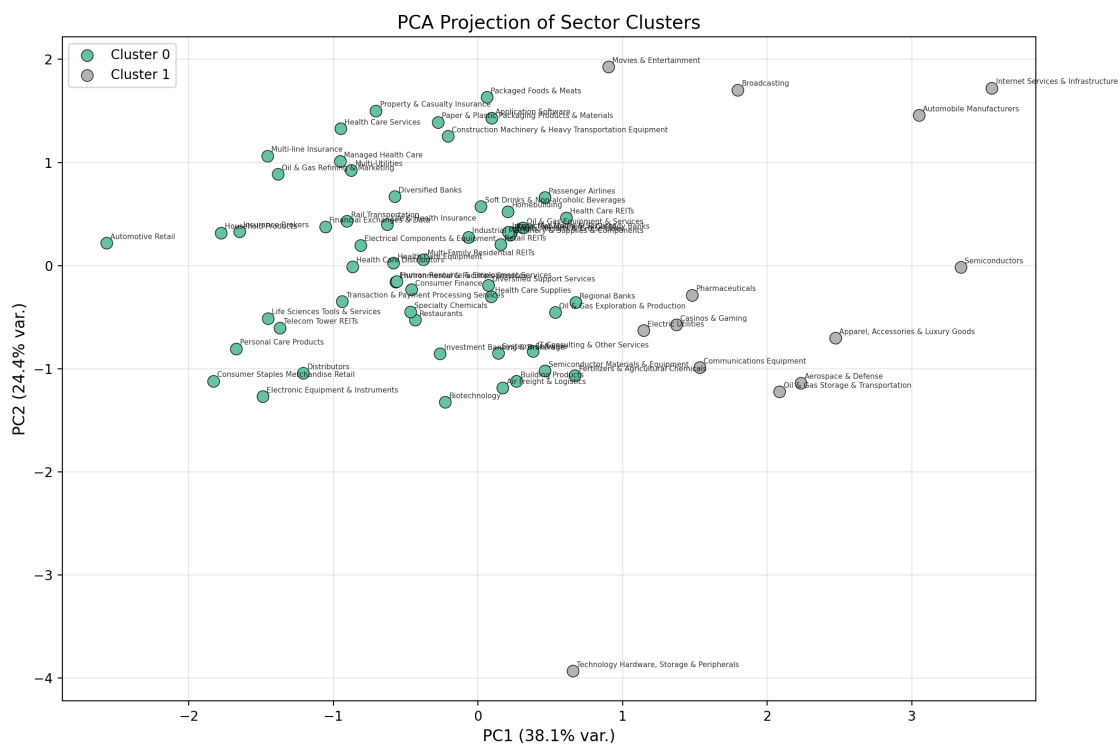


Figure 5: PCA projection of sector clusters. The first two principal components capture the dominant variance structure, with high-concentration outlier sectors (Cluster 1) clearly separated from the main group.

4.3.3 Classification Validation

Leave-one-out cross-validation with a Random Forest classifier achieves **95.6%** accuracy, substantially above the 50% expected by chance for 2 classes. Only 3 of 68 sectors are misclassified. Table 2 presents the full validation metrics.

Table 2: Clustering Validation Metrics

Metric	Score
Silhouette Score	0.360
Calinski-Harabasz Index	22.01
Davies-Bouldin Index	1.352
LOO-CV Accuracy	95.6%
Adjusted Rand Index	0.791
Normalized Mutual Info.	0.694
V-Measure	0.694
Fowlkes-Mallows Index	0.941

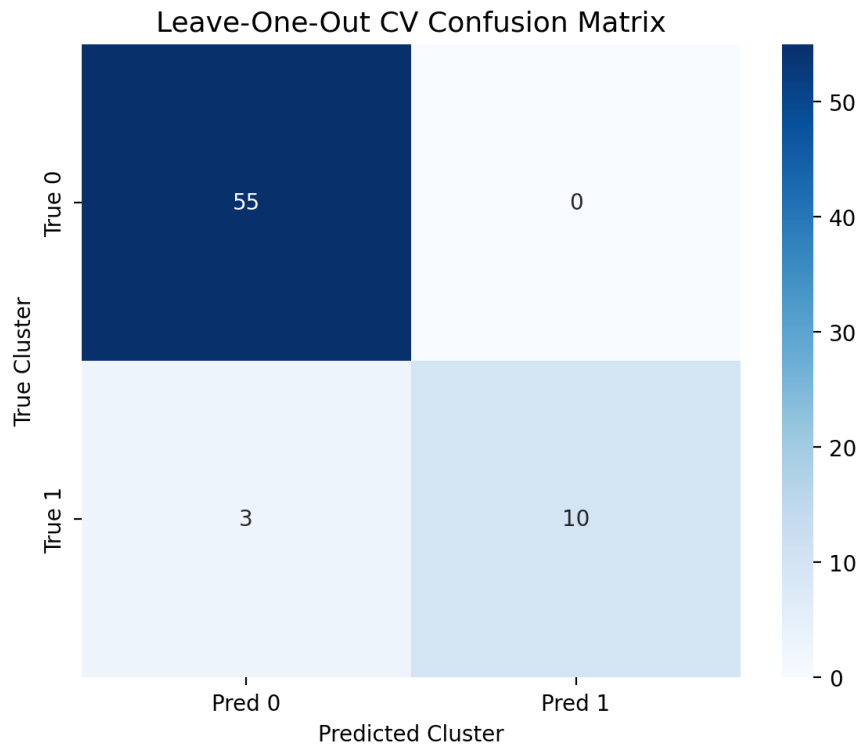


Figure 6: Leave-one-out cross-validation confusion matrix. All 55 Cluster 0 sectors are correctly classified; 3 of 13 Cluster 1 sectors are misclassified as Cluster 0, representing borderline cases at the concentration threshold.

4.4 In-Sample / Out-of-Sample Validation

To test whether market structures are persistent, we train on 2019-2023 (in-sample, 17 quarters) and evaluate on 2024-2025 (out-of-sample, 8 quarters). Two key findings emerge:

Market structure classifications are moderately stable: 75.0% of sectors (51/68) receive the same market structure label (monopolistic, duopolistic, or oligopolistic) in both the in-sample and out-of-sample periods. This suggests that the competitive structure of most sectors persists across a 2-year horizon, though a quarter of sectors undergo structural transitions.

Overperformance rankings reshuffle significantly: The Pearson correlation between in-sample and out-of-sample overperformance index values is $r = 0.181$ ($p = 0.14$), which is not statistically significant. This is a central finding: while the *type* of market structure (monopolistic vs. oligopolistic) tends to persist, the *identity* of the dominant firm within a sector changes substantially between periods.

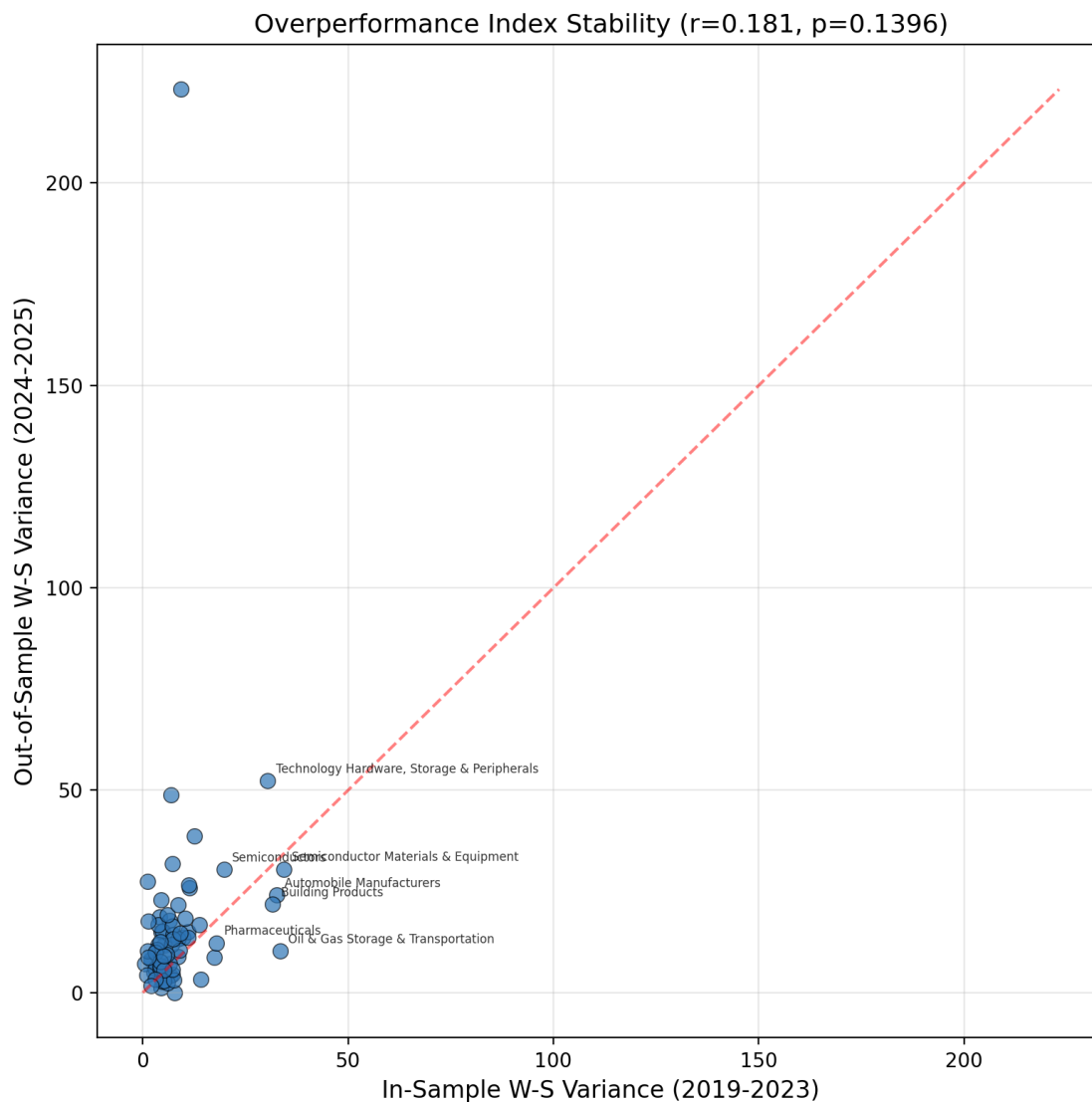


Figure 7: Overperformance index: in-sample (2019-2023) vs. out-of-sample (2024-2025). The weak correlation ($r = 0.181$, $p = 0.14$) indicates that which firms dominate a sector's growth is not persistent, even when the overall market structure is.

This result has practical implications for portfolio construction: sector-level allocation based on market structure may be stable over multi-year horizons, but stock selection within sectors requires

frequent reassessment as dominant firms rotate.

4.5 Time-Varying Market Structure

Rolling-window analysis with an 8-quarter (2-year) window confirms that market concentration is not static. Fig. 8 shows the rolling HHI for the six most concentrated sectors. Several patterns emerge:

- Technology Hardware maintains consistently extreme concentration throughout the full period, reflecting Apple’s persistent dominance.
- Oil & Gas Storage & Transportation shows a marked increase in concentration during 2021-2022, coinciding with the energy price shock, followed by partial reversion.
- Aerospace & Defense displays cyclical concentration patterns, suggesting competitive dynamics shift with defense spending cycles.
- Internet Services & Infrastructure concentration spiked during 2024-2025, driven by Palantir’s extraordinary market cap growth during the AI rally.

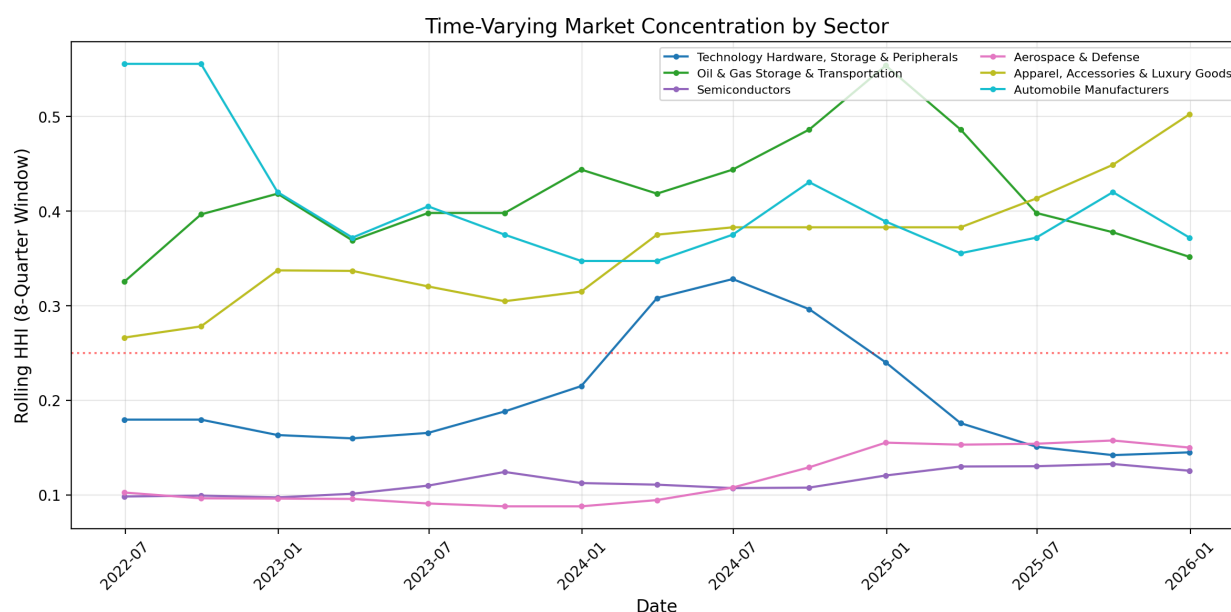


Figure 8: Rolling 8-quarter HHI for the six most concentrated sectors. The horizontal dashed line at $HHI = 0.25$ marks the DOJ threshold for high concentration. Market structures are visibly non-stationary, with concentration levels shifting in response to macroeconomic and sector-specific shocks.

4.6 Cross-Sector Connectedness

An earlier version of this study reported “297 significant Granger pairs” from pairwise tests on 25 quarterly observations at an uncorrected $p < 0.05$. That count is a multiple-comparisons artifact: across $68 \times 67 = 4,556$ directed pairs roughly 235 are significant by chance alone, and *zero* survive Benjamini-Hochberg FDR or Bonferroni correction. We replace it with two properly powered analyses on daily market-cap-weighted sector returns (1,590 observations across the 11 GICS sectors).

Generalized connectedness. Following Diebold and Yilmaz (2012), we fit a VAR (lag selected by AIC) and compute the generalized forecast-error variance decomposition (Pesaran and Shin, 1998) at a 10-day horizon, row-normalized into a connectedness table. The total connectedness index is **78.6%**: most of each sector’s forecast-error variance is explained by shocks to *other* sectors. Net directional connectedness (Table 3) ranks sectors as transmitters or receivers.

Table 3: Net Directional Connectedness (transmitter > 0 > receiver)

Transmitter	NET	Receiver	NET
Industrials	+24.2	Energy	−23.4
Financials	+24.1	Communication Services	−14.9
Materials	+15.9	Utilities	−10.4
Health Care	+3.5	Consumer Discretionary	−10.4
Real Estate	+0.1	Consumer Staples	−8.6

Cyclical hubs (Industrials, Financials, Materials) propagate shocks system-wide, while Energy is the largest net receiver, consistent with its idiosyncratic, commodity-driven dynamics. Fig. 9 visualizes the directed network implied by the connectedness table, and Fig. 10 the net positions. A rolling 200-day connectedness index (Fig. 11) ranges from 55.5% to a peak of 87.0% on 24 June 2020, tracing systemic linkage through the COVID-19 crisis.

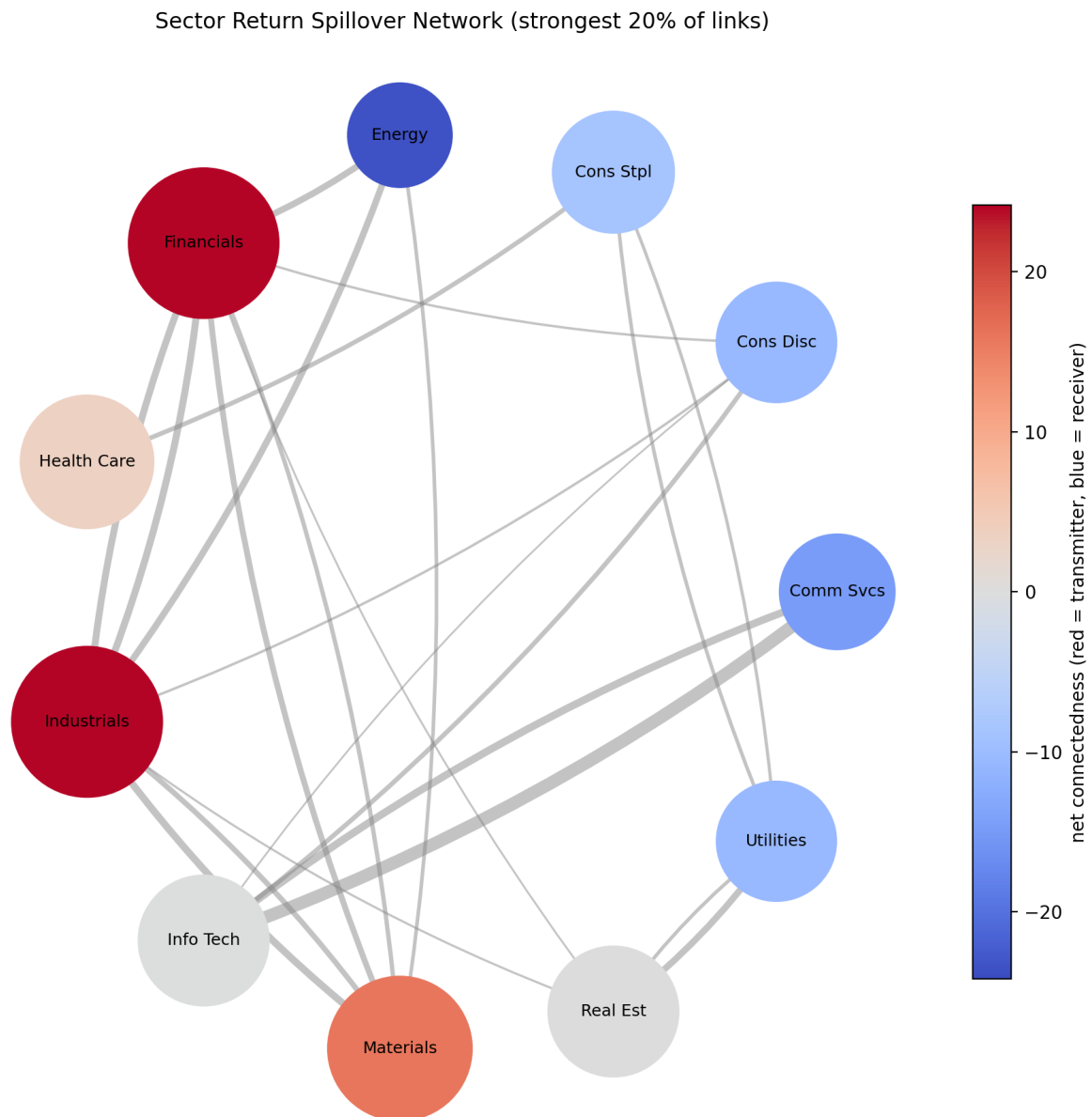


Figure 9: Directed spillover network from the generalized variance decomposition (10-day horizon). Edge intensity is pairwise directional connectedness; node color encodes the net transmitter (warm) versus receiver (cool) position. The cyclical hubs sit at the center of the transmission structure.

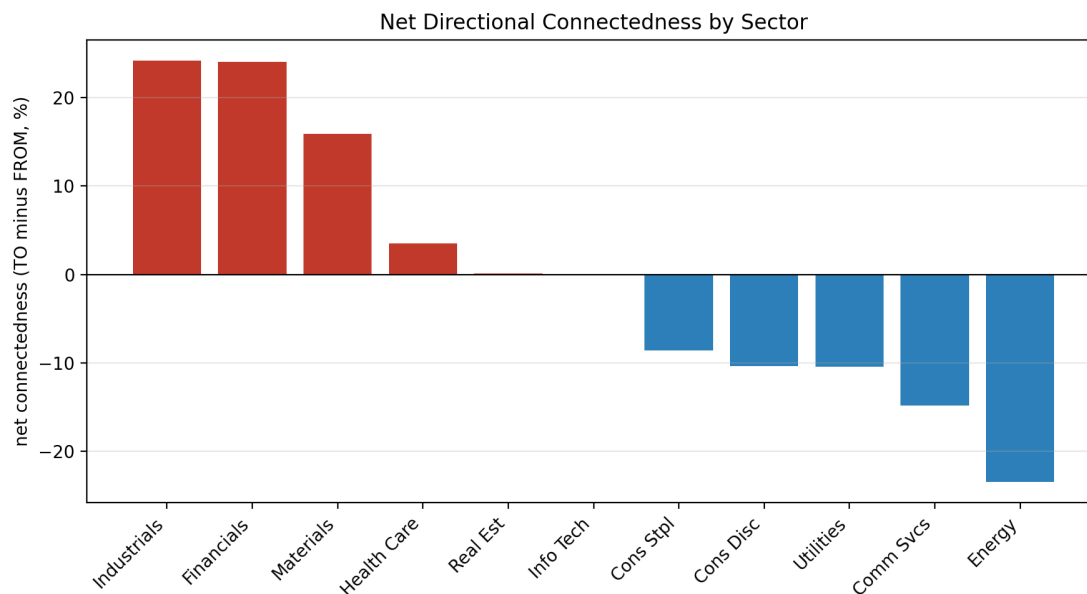


Figure 10: Net directional connectedness by sector (transmitters positive, receivers negative), the graphical counterpart of Table 3.

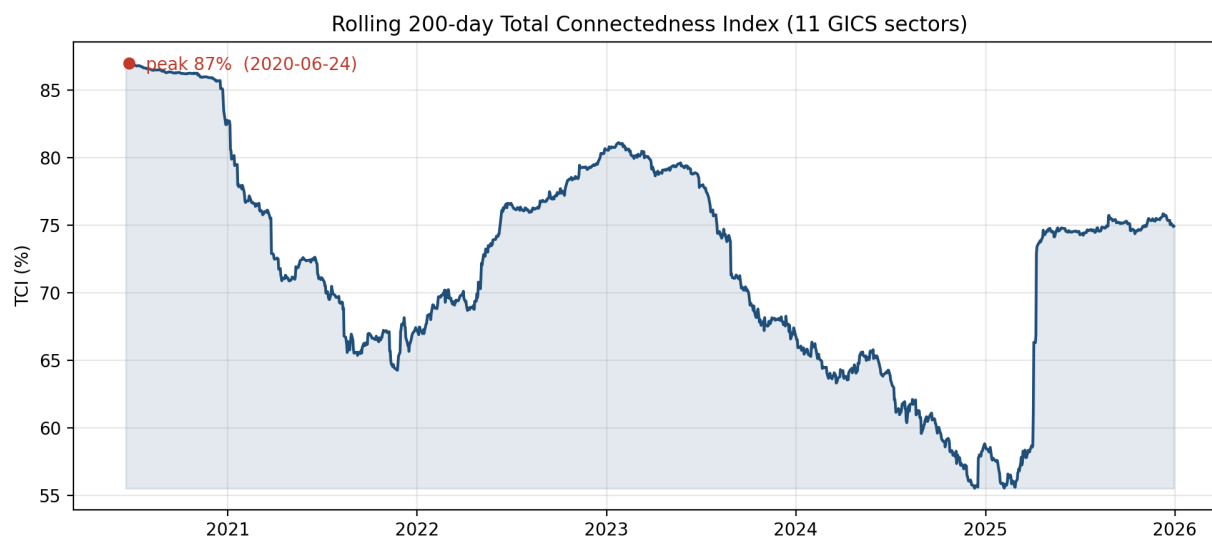


Figure 11: Rolling 200-day total connectedness index of daily sector returns. Systemic linkage jumps in the COVID-19 crash (peak 87.0% on 24 June 2020) and never returns to its pre-2020 lows, consistent with a persistently more connected sector system.

Granger causality, done right. On the daily series with maximum lag 5 and multiple-testing correction, 88 of 110 directed links are significant under FDR and 54 under the stricter Bonferroni (versus about 6 expected by chance); Fig. 12 maps the surviving links. The contrast with the quarterly result (0 surviving correction) is the methodological point: sampling frequency and multiple-testing control determine whether cross-sector causality is detectable at all.

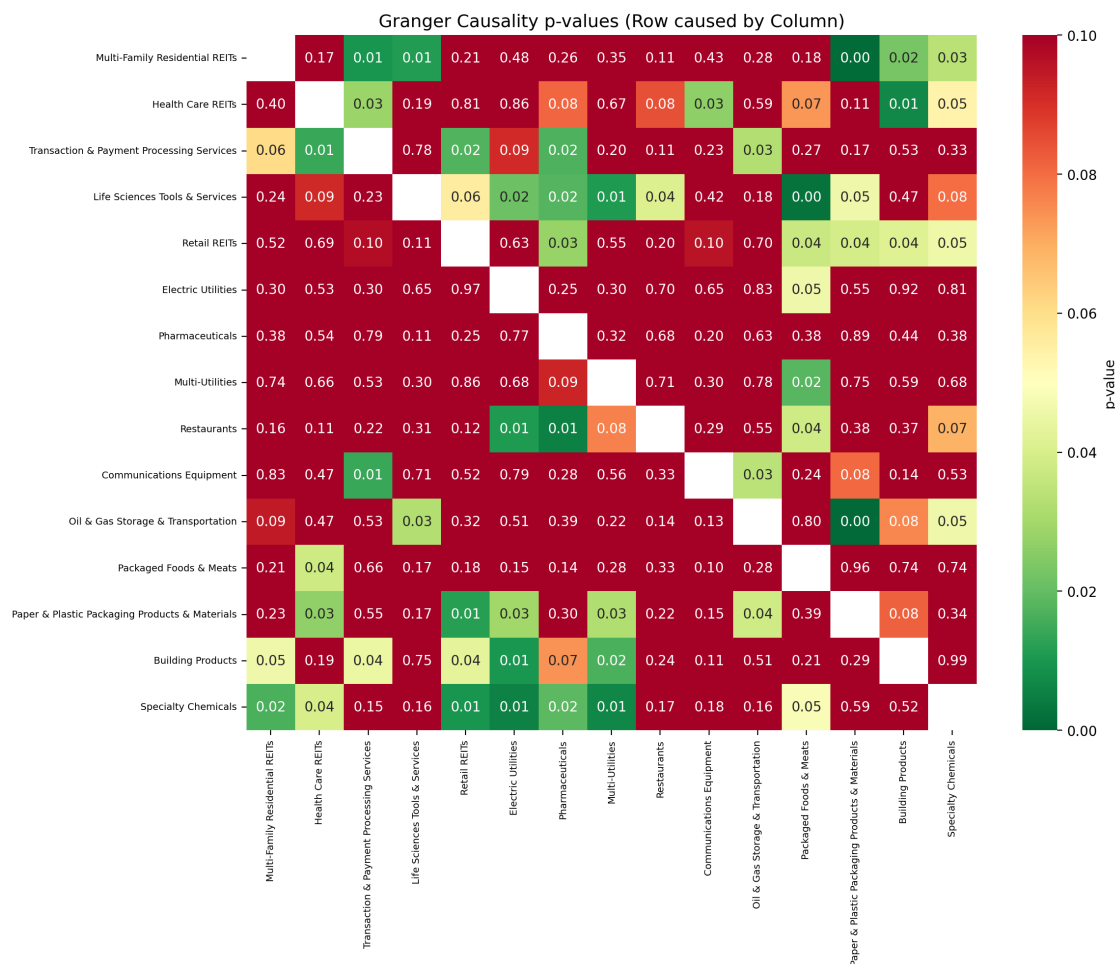


Figure 12: Pairwise Granger causality on daily sector returns (max lag 5). Cells mark directed links significant after Benjamini-Hochberg FDR control (88 of 110; 54 survive Bonferroni). The same test on 25 quarterly observations yields zero links after correction.

4.7 Reproducibility

Every number and figure in this paper regenerates from the public repository with one command per stage: `python src/build_sector_returns.py` (constituent prices to daily market-cap-weighted GICS sector returns), `python src/connectedness.py` (VAR, generalized FEVD, rolling index, corrected Granger tests), `python src/hhi_dynamics.py` (concentration dynamics), and `python src/analysis.py` (clustering and market-structure classification). Outputs are written to `output/` as CSV tables and PNG figures consumed directly by this document.

5 Discussion

5.1 Methodological Contributions

This study makes three methodological contributions to sector-level financial analysis.

First, the overperformance index with squared-weight allocation provides a growth-based concentration metric that captures dynamic competitive advantages invisible to static HHI. The significant correlation ($r = 0.537$) with traditional HHI validates the metric while the imperfect agreement

demonstrates its complementary value.

Second, the use of binomial significance tests replaces arbitrary thresholds for market structure classification. Under the null hypothesis that outperformance is random, we can formally reject the null for genuinely dominant firms while avoiding false positives from sampling noise.

Third, the temporal out-of-sample validation reveals an important asymmetry: market *structure* classifications are stable (75% agreement across a 2-year holdout), but the *identity* of dominant firms reshuffles significantly ($r = 0.181$). This implies that sector-level allocation strategies based on market structure typology are more robust than within-sector stock picks based on historical dominance patterns.

5.2 Practical Implications

The finding that market structures are moderately persistent but dominant firms rotate has direct implications for portfolio construction:

- **Sector allocation:** The binary cluster structure (concentrated vs. dispersed sectors) provides a stable framework for sector-level risk budgeting. Concentrated sectors carry higher idiosyncratic risk due to single-stock dependence.
- **Stock selection:** Within-sector stock selection based on past overperformance is not a reliable strategy, as the $r = 0.181$ IS/OOS correlation shows. Factor-based approaches within sectors may be more robust.
- **Sector rotation:** high daily connectedness (TCI 78.6%) and 88 of 110 FDR-significant Granger links indicate cross-sector return information is pervasive, so lead-lag and sector-momentum signals plausibly carry predictive content; the net-transmitter ranking (Industrials, Financials) points to where shocks originate.

5.3 Limitations

Several limitations should be acknowledged:

- **Shares outstanding:** Market cap is computed using current shares outstanding multiplied by historical Close prices. This does not account for historical share issuance or buybacks, introducing a known upward bias in historical market caps for companies with net buybacks.
- **Sample size:** With 68 sectors and 5 features, the clustering analysis operates in a regime where sample size is modest relative to dimensionality. The use of LOO-CV rather than neural networks is an explicit design choice to respect this constraint.
- **Survivorship bias:** The analysis uses current S&P 500 constituents, which may not reflect the competitive landscape during earlier periods when different companies were included.
- **GICS classification:** Sub-industry boundaries are somewhat arbitrary. Companies like Amazon span multiple functional sectors.
- **Granger $\neq$ causation:** Lead-lag relationships indicate predictive information flow, not economic causation. Omitted common factors may drive apparent inter-sector dependencies.

6 Conclusion

This paper provides a rigorous, multi-method analysis of competitive dynamics within S&P 500 sub-sectors using data through December 2025. Key findings include:

1. **Market structures are heterogeneous:** 37 of 68 sectors (54%) exhibit oligopolistic competition, while 22 (32%) show statistically significant monopolistic tendencies. The extreme outliers are Technology Hardware (Apple: 94.1%), Automobile Manufacturers (Tesla: 93.1%), and Internet Services (Palantir: 88.8%, reflecting the 2024-2025 AI rally).
2. **Growth dominance partially tracks size dominance:** The overperformance HHI correlates with market-cap HHI at $r = 0.537$ ($p < 10^{-4}$), validating the overperformance index while revealing that growth-based and size-based concentration provide complementary views.
3. **Two-cluster structure:** Silhouette-optimized clustering identifies a clean binary partition (score = 0.360, LOO-CV = 95.6%, FM = 0.941): 55 sectors with moderate, dispersed competition and 13 sectors with extreme concentration.
4. **Market structures are stable but dominant firms rotate:** 75% of sectors maintain their classification out-of-sample, but overperformance rankings reshuffle significantly ($r = 0.181$, $p = 0.14$). This asymmetry implies that sector-level allocation is more robust than within-sector stock selection based on past dominance.
5. **Sectors are highly interdependent:** a Diebold-Yilmaz total connectedness index of 78.6% and 88 of 110 FDR-significant daily Granger links establish pervasive cross-sector transmission, with Industrials and Financials the dominant net transmitters and Energy the dominant net receiver; the same tests on quarterly data detect nothing after correction.

Future work should extend the window to a full business cycle, estimate time-varying connectedness with a rolling or TVP-VAR to study how transmitter and receiver roles rotate across regimes, incorporate ESG metrics as additional clustering features, and test whether the connectedness network and sector clusters predict differential risk-adjusted returns in an out-of-sample allocation framework.

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